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WATERFALL DECKPLATE

Abstract

A waterfall deckplate assembly that includes a deckplate structure adapted to be mounted on a horizontal surface and a fluid conveyance system coupled with deckplate structure. The fluid conveyance system is adapted to be connected to a fluid supply and includes a plurality of discharge nozzles supported by the deckplate structure wherein the plurality of discharge nozzles are horizontally aligned when the deckplate structure is mounted on the horizontal surface; and a user operable control valve wherein the control valve is operable to independently and selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles. In some embodiments, the plurality of nozzles are formed on a first nozzle manifold and the deckplate assembly includes a storage compartment for a second nozzle manifold having a second plurality of discharge nozzles.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. 119(e) of U.S. provisional patent application Ser. No. 63/556,974 filed on Feb. 23, 2024 and U.S. provisional patent application Ser. No. 63/650,548 filed on May 22, 2024, the disclosures of both of which are hereby incorporated herein by reference.

BACKGROUND AND SUMMARY OF THE DISCLOSURE
[0002] The present disclosure relates to a fluid dispensing device and, more particularly, to a waterfall deckplate for dispensing water.
[0003] Waterfall fluid dispensers are known in the art. For example, it is known to combine waterfall dispensers with a kitchen faucet whereby fluid can be dispensed from either the waterfall dispenser or the faucet into a kitchen sink. In such an application, the waterfall dispenser can beneficially be used to wash or rinse fresh produce and other food.
[0004] While known dispensers are useful, further improvements remain desirable.
[0005] The present disclosure discloses a waterfall deckplate assembly which includes a deckplate structure adapted to be mounted on a horizontal surface, and a fluid conveyance system coupled with deckplate structure, the fluid conveyance system adapted to be connected to a fluid supply and including: a plurality of discharge nozzles supported by the deckplate structure wherein the plurality of discharge nozzles are horizontally aligned when the deckplate structure is mounted on the horizontal surface; and a user operable control valve wherein the control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles.
[0006] In some embodiments of the waterfall deckplate assembly, the control valve is a positioned downstream of a mixing valve adapted to be coupled with a hot water supply and a cold water supply and a diverter valve is disposed between the mixing valve and the control valve, the diverter valve being couplable with a faucet. The faucet may be mounted on the deckplate assembly or mounted separately and spaced apart from the deckplate assembly.
[0007] In some embodiments of the waterfall deckplate assembly, the plurality of discharge nozzles are adjustable nozzles.
[0008] In some embodiments of the waterfall deckplate assembly, the deckplate structure defines an opening adapted to receive a faucet assembly. The opening is, in some embodiments, centrally located on the deckplate assembly.
[0009] In some embodiments of the waterfall deckplate assembly, the control valve includes a user operable actuator projecting above the deckplate structure for operating the control valve. In some embodiments, there are a plurality of user operable actuators interchangeably installable on the control valve wherein each of the plurality of user operable actuators has a different exterior appearance.
[0010] In some embodiments of the waterfall deckplate assembly, the plurality of discharge nozzles are disposed on a first nozzle manifold and the assembly includes a second plurality of discharge nozzles disposed on a second nozzle manifold, the first and second nozzle manifolds being interchangeably installable in the fluid conveyance system; and wherein the deckplate structure defines a storage compartment whereby one of the first and second nozzle manifolds not installed in the fluid conveyance system is storable in the storage compartment.
[0011] In some embodiments of the waterfall deckplate assembly, the plurality of discharge nozzles are disposed on a nozzle manifold wherein the nozzle manifold includes at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and the

assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.

[0012] In some embodiments of the waterfall deckplate assembly, the control valve is a mixing valve adapted to be coupled with a hot water supply and a cold water supply.

[0013] In some embodiments of the waterfall deckplate assembly, the waterfall deckplate assembly is adapted to be mounted proximate a faucet whereby both the waterfall deckplate assembly and the faucet discharge fluid into a common basin with the faucet being mounted separately and spaced apart from the deckplate assembly.

[0014] Another embodiment takes the form of a waterfall deckplate assembly that includes a deckplate structure adapted to be mounted on a horizontal surface; a first plurality of discharge nozzles disposed on a first nozzle manifold and a second plurality of discharge nozzles disposed on a second nozzle manifold; a fluid conveyance system coupled with deckplate structure, the fluid conveyance system being couplable with a selected one of the first nozzle manifold and the second nozzle manifold, and wherein the fluid conveyance system is connectable to a fluid supply to thereby convey fluid to the selected one of the first nozzle manifold and the second nozzle manifold coupled with the fluid conveyance system; a user operable control valve disposed in the fluid conveyance system wherein the control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles; and wherein the first nozzle manifold and the second nozzle manifold are interchangeably couplable to the fluid conveyance system and the deckplate structure defines a storage compartment whereby a non-coupled one of the first nozzle manifold and the second nozzle manifold is storable in the storage compartment.

[0015] In some embodiments of the waterfall deckplate assembly, the plurality of discharge nozzles of the nozzle manifold coupled with the fluid conveyance system are horizontally aligned when the deckplate structure is mounted on the horizontal surface.

[0016] In some embodiments of the waterfall deckplate assembly, the storage compartment is disposed below the nozzle manifold coupled with the fluid conveyance system when the deckplate structure is mounted on the horizontal surface. In some embodiments, the nozzle manifold coupled with the fluid conveyance system discharges fluid from a first elongate side of the deckplate structure and the storage compartment is accessible through a removable cover disposed on a second elongate side of the deckplate structure disposed opposite the first elongate side.

[0017] In some embodiments of the waterfall deckplate assembly, at least one of the first plurality of discharge nozzles and the second plurality of discharge nozzles are adjustable nozzles.

[0018] In some embodiments of the waterfall deckplate assembly, the plurality of discharge nozzles are disposed on a nozzle manifold wherein the nozzle manifold includes at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.

[0019] Another embodiment takes the form of a faucet assembly that includes a faucet having a waterway connectable to a water supply, the waterway extending to a discharge outlet, and a user operable first control valve for controlling the flow of water through the waterway to the discharge outlet; and a deckplate assembly that includes: a deckplate structure adapted to be mounted on a horizontal surface; and a fluid conveyance system coupled with deckplate structure, the fluid conveyance system adapted to be connected to a fluid supply and including: a plurality of discharge nozzles supported by the deckplate structure wherein the plurality of discharge nozzles are horizontally aligned when the deckplate structure is mounted on the horizontal surface; and a user operable second control valve wherein the second control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles independently of the operation of the first control valve.

[0020] In some embodiments of the faucet assembly, the first control valve is a mixing valve adapted to be coupled with a hot water supply and a cold water supply and a diverter valve is disposed between the mixing valve and the second control valve, with both the waterway and the fluid conveyance system being disposed downstream of the diverter valve and the mixing valve being disposed upstream of the diverter valve.

[0021] In some embodiments of the faucet assembly, the second control valve includes a user operable actuator projecting above the deckplate structure for operating the second control valve. The faucet assembly may also include a plurality of user operable actuators interchangeably installable on the second control valve, each of the plurality of user operable actuators having a different exterior appearance.

[0022] In some embodiments of the faucet assembly, the plurality of discharge nozzles are disposed on a first nozzle manifold and the assembly includes a second plurality of discharge nozzles disposed on a second nozzle manifold, the first and second nozzle manifolds being interchangeably installable in the fluid conveyance system; and wherein the deckplate structure defines a storage compartment whereby one of the first and second nozzle manifolds not installed in the fluid conveyance system is storable in the storage compartment.

[0023] In some embodiments of the faucet assembly, the plurality of discharge nozzles are disposed on a nozzle manifold wherein the nozzle manifold includes at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.

[0024] In some embodiments of the faucet assembly the deckplate structure defines an opening and the faucet is mounted on the deckplate structure at the opening.

[0025] In some embodiments of the faucet assembly the waterfall deckplate assembly is adapted to be mounted proximate the faucet whereby both the waterfall deckplate assembly and the faucet discharge fluid into a common basin with the faucet being mounted separately and spaced apart from the deckplate assembly.

[0026] Yet another embodiment takes the form of a faucet assembly that includes a faucet having a waterway connectable to a water supply, the waterway extending to a discharge outlet, and a user operable first control valve for controlling the flow of water through the waterway to the discharge outlet; and a deckplate assembly that includes: a deckplate structure adapted to be mounted on a horizontal surface; a first plurality of discharge nozzles disposed on a first nozzle manifold and a second plurality of discharge nozzles disposed on a second nozzle manifold; a fluid conveyance system coupled with deckplate structure, the fluid conveyance system being couplable with a selected one of the first nozzle manifold and the second nozzle manifold, and wherein the fluid conveyance system is connectable to a fluid supply to thereby convey fluid to the selected one of the first nozzle manifold and the second nozzle manifold coupled with the fluid conveyance system; a user operable control valve disposed in the fluid conveyance system wherein the control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles; and wherein the first nozzle manifold and the second nozzle manifold are interchangeably couplable to the fluid conveyance system and the deckplate structure defines a storage compartment whereby a non-coupled one of the first nozzle manifold and the second nozzle manifold is storable in the storage compartment.

[0027] In some embodiments of the faucet assembly, the plurality of discharge nozzles of the nozzle manifold coupled with the fluid conveyance system are horizontally aligned when the deckplate structure is mounted on the horizontal surface.

[0028] In some embodiments of the faucet assembly, the storage compartment is disposed below the nozzle manifold coupled with the fluid conveyance system when the deckplate structure is mounted on the horizontal surface. The faucet assembly may be configured such that the nozzle manifold coupled with the fluid conveyance system discharges fluid from a first elongate side of the deckplate structure and the storage compartment is accessible through a removable cover disposed on a second elongate side of the deckplate structure disposed opposite the first elongate side.

[0029] In some embodiments of the faucet assembly, the plurality of discharge nozzles are disposed on a nozzle manifold wherein the nozzle manifold includes at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.

[0030] Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] A detailed description of the drawings particularly refers to the accompanying figures, in which:

[0032] FIG. **1**A is a perspective view of an illustrative waterfall deckplate if the present disclosure supported by a sink basin and operably coupled to a faucet;

[0033] FIG. **1**B is a schematic view of the illustrative waterfall deckplate of FIG. **1**A;

[0034] FIG. **2** is a front perspective view of the illustrative waterfall deckplate of FIG. **1**A;

[0035] FIG. **3** is a rear perspective view of the illustrative waterfall deckplate of FIG. **2**;

[0036] FIG. **4** is an exploded perspective view of the illustrative waterfall deckplate of FIG. **2**;

[0037] FIG. **5** is a cross-sectional view of the illustrative waterfall deckplate taken along line **5-5** of FIG. **2**;

[0038] FIG. **6** is a cross-sectional view of the illustrative waterfall deckplate taken along line **6-6** of FIG. **2**;

[0039] FIG. **7** is a cross-sectional view of the illustrative waterfall deckplate taken along line **7-7** of FIG. **2**;

[0040] FIG. **8** is a front perspective view of another illustrative waterfall deckplate;

[0041] FIG. **9** is a rear perspective view of the illustrative waterfall deckplate of FIG. **8**;

[0042] FIG. **10** is an exploded perspective view of the illustrative waterfall deckplate of FIG. **8**;

[0043] FIG. **11**A is an exploded perspective view of the fluid discharge assembly of the illustrative waterfall deckplate of FIG. **8**;

[0044] FIG. **11**B is an exploded view showing a manifold assembly and cover plate of FIG. **8**;

[0045] FIG. **12**A is a cross-sectional view of the illustrative waterfall deckplate taken along line **12**A-**12**A of FIG. **8**;

[0046] FIG. **12**B is a detail view of a portion of FIG. **12**A;

[0047] FIG. **13** is a cross-sectional view of the illustrative waterfall deckplate taken along line **13-13** of FIG. **8**; and

[0048] FIG. **14** are perspective views of various control knobs for use with the illustrative waterfall deckplates illustrated herein.

[0049] FIG. **15** is a perspective view of another illustrative waterfall deckplate assembly and faucet.

[0050] FIG. **16** is a rear perspective view of the deckplate assembly of FIG. **15**.

[0051] FIG. **17** is an exploded view of the deckplate assembly of FIG. **15**.

[0052] FIG. **18** is a cross-sectional view taken along line **18-18** of FIG. **16**.

[0053] FIG. **19** is a cross-sectional view taken along line **19-19** of FIG. **16**.
[0054] FIG. **20** is a cross-sectional view taken along line **20-20** of FIG. **16**.
[0055] FIG. **21** is a cross-sectional view taken along line **21-21** of FIG. **16**.
[0056] FIG. **22** is schematic perspective view of an upper valve body of the waterfall deckplate assembly of FIG. **15**.
[0057] FIG. **23** is a schematic perspective view of a lower valve body of the waterfall deckplate assembly of FIG. **15**.
[0058] Corresponding reference characters indicate corresponding parts throughout the several views.

DETAILED DESCRIPTION OF THE DRAWINGS

[0059] For the purposes of promoting and understanding the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, which are described herein. The embodiments disclosed herein are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings. Therefore, no limitation of the scope of the claimed invention is thereby intended. The present invention includes any alterations and further modifications of the illustrated devices and described methods and further applications of principles in the invention which would normally occur to one skilled in the art to which the invention relates.
[0060] A faucet assembly **20** is shown in FIG. **1**A and includes a faucet **22** and a waterfall deckplate assembly **30**. In the illustrated embodiment, faucet **22** is a conventional spring faucet having a spring **24** that surrounds a flexible portion **26** and a discharge outlet **28**, illustratively defined by a movable sprayhead **29**. Other forms of faucets may also be employed with deckplate assembly **30**.
[0061] In FIG. **1**A, deckplate assembly **30** is shown mounted on a sink deck or rim **32** of a kitchen sink whereby the faucet assembly **20** may discharge fluid, e.g., water, into the basin **34** of the kitchen sink. Rim **32** provides a substantially flat horizontal surface for mounting deckplate assembly **30**, however, deckplate assembly **30** may also be mounted on alternative surfaces. For example, deckplate assembly **30** could be mounted on the upper surface of a countertop positioned adjacent a sink basin. Advantageously, the surface on which the deckplate assembly is mounted is a substantially flat and horizontal surface.
[0062] Deckplate assembly **30** is illustrated in FIGS. **1**A-**7** and includes a deckplate structure **36** adapted to be mounted on a horizontal surface. The illustrative deckplate structure **36** includes a framework **38** and a cover plate **40**. Cover plate **40** includes a plurality of prongs **41** that are used to provide a snap-fit connection between cover plate **40** and framework **38**, however, alternative attachment methods may also be employed. Hollow attachment posts or mounting shanks **42**, **44** are secured to framework **38** and extend downwardly through openings in a horizontal supporting member when deckplate assembly **30** is mounted on the upper surface of the horizontal supporting member. In the illustrated embodiments, posts **42**, **44** are externally threaded and threaded nuts **46** are threaded onto posts **42**, **44** to engage the underside of the horizontal supporting member and thereby secure the deckplate assembly to the horizontal supporting member, e.g., the rim of a sink or a countertop. Post **42** is also threadingly engaged with a threaded bore in framework **38** to secure the post **42** to framework **38**. Alternative configurations may also be employed, e.g., post **42** may be integrally formed with framework **38** or be secured to framework **38**. Furthermore, a retaining member on the underside of the horizontal supporting member using a different attachment method, e.g., friction fit, interference fit, bayonet coupling, adhesive and/or some other suitable attachment method may alternatively be used with posts **42**, **44**.
[0063] In the illustrated embodiment, post **44** is a hollow member having a central bore that extends the entire length of post **44**. Post **44** also includes an upper section **48** in which control valve **50** is positioned. A lateral, hollow, cylindrical extension **52** extends outwardly from upper

section **48**. Extension **52** includes two circumferential grooves on its cylindrical exterior surface for seating O-ring seals **54**. A fluid line **104** conveys fluid to control valve **50** and, when control valve **50** is open, the fluid is allowed to flow through extension **52** to manifold assembly **60**. See FIG. **11**A for an exploded view of manifold assembly **160**. As further discussed below, manifold assembly **60** is identical to manifold assembly **160** except that the manifold housing **172** of manifold assembly **160** includes a pair of brackets **174** on the rear wall of manifold housing **172**.
[0064] Post **44** includes lateral tabs **62** that are engaged in a slot and rest on a shelf in framework **38** such that tightening nut **46** on post **44** secures framework **38** to the horizontal supporting member by the compressive forces between tabs **62** and nut **46**. Threaded fasteners (not shown) are used to firmly secure tabs **62** to framework **38**. It is noted that upper section **48** can be integrally formed with post **44** or secured thereto. In the illustrated embodiment, upper section **48** is threadingly secured to post **44**. After attaching upper section **48** to post **44**, the combined assembly is secured to framework **38** via tabs **62** and threaded fasteners. In other embodiments, the section that is used to house control valve **50** could be entirely separate from the structural member used to secure deckplate assembly **30** to the horizontal supporting member.
[0065] A nut **58** is used to secure control valve **50** within upper section **48**. Opposing slots in the upper edge of upper section **48** are used to engage lateral projections on control valve **50** to prevent relative rotation of the valve housing with upper section **48**. A user operable actuator **64** or knob is coupled with control valve **50**, e.g., by engaging a stem on valve **50**, whereby a user may rotate knob **64** to open and close valve **50**. Valve **50** is a conventional fluid valve such as a ball valve, diaphragm valve or other suitable fluid valve.
[0066] Opening valve **50** allows fluid to flow to extension **52** and then to nozzle manifold **60**. Closing valve **50** prevents the flow of fluid to extension **52** which thereby prevents the flow of fluid to nozzle manifold **60**. Extension **52** is sealingly received in end cap **66** of nozzle manifold **60**. End cap **66** has a tubular extension **68** similar to extension **52** on which O-ring seals **70** are seated. End cap **66** defines an internal passageway that is in fluid communication with the interior bore of extension **52** and extends through tubular extension **68**. Tubular extension **68** is received in manifold body **76** whereby fluid can flow from control valve **50**, through extension **52** and end cap **66** into manifold body **76**. Manifold body **76** is disposed in manifold housing **72** which, in turn, is supported on framework **38**. An opposing end cap **74** is engaged with the capped end of manifold body **76** which is opposite end cap **66**. Opposing end cap **74** has a configuration identical to end cap **66**, however, manifold body has an opening allowing end cap **66** to communicate with internal passage **78** while the other end of passage **78** is a blind bore whereby fluid cannot flow into end cap **74**. A short blind bore having a depth equivalent to tubular extension **68** receives the tubular extension **68** of end cap **74**. In addition to internal passage **78**, manifold body **76** also defines a plurality of discharge nozzles **80**. Fluid entering passage **78** from end cap **66** is discharged from nozzles **80** to form a waterfall discharge. In this regard, it is noted that the discharge nozzles **80** are horizontally aligned when the deckplate structure is mounted on the horizontal surface to thereby form a waterfall discharge. When valve **50** is open, fluid is allowed to flow to discharge nozzles **80**, it will be discharged from nozzles in the form of a waterfall and when valve **50** is closed, fluid is prevented from flowing to discharge nozzles **80** and the fluid flow from nozzles **80** will thereby be terminated.
[0067] In other words, valve **50** toggles the water flow on and off through the fluid conveyance system **31** of the waterfall deckplate assembly. Fluid conveyance system **31** includes all those elements of the waterfall deckplate assembly that control and convey fluid through the assembly which, in waterfall deckplate assembly **30**, include valve **50**, extension **52** and manifold assembly **60**. The fluid is discharged through the plurality of discharge nozzles **80** which are a part of manifold assembly **60** and also a part of the fluid conveyance system **31**.
[0068] In the illustrated embodiment, manifold body **76** also includes a pair of projecting tabs **82** that a user can engage to rotate manifold body **76** within manifold housing **72** and about the tubular

projections of end caps **66**, **74**. By rotating manifold body **76**, the orientation of nozzles **80** can be adjusted whereby the angle at which the fluid being discharged can be adjusted, i.e., directed in a more upward or downward direction depending upon the position of manifold body **76** within manifold housing **72**. As shown in FIG. **11**A, nozzles **80** may be formed in an inlay **84** that is subsequently attached to manifold body **76**, e.g., by friction fit, adhesives or other suitable method.

[0069] Latching members **86** are used to secure manifold assembly **60** to framework **38**. Latching members **86** include a pair of resiliently flexible barbed tabs **110** wherein barbs **111** extend outwardly from the tabs **110** to engage a ledge **39** on framework **38** whereby latching members **86** can be pushed further into framework **38** but cannot be removed without biasing the barbed tabs **110** inwardly. Latching members **86** also include a camming member **112** for releasing manifold housing **72** from framework **38**. Two pair of resiliently flexible barbed prongs **114** extend from the back wall of the manifold housing wherein the barbs project outwardly from the tabs and engage ledges **37** on the framework to thereby retain the manifold assembly on the framework. As best understood with reference FIGS. **12**A and **12**B, depression of latching members **86** causes the camming member **112** on a latching member **86** to bias a pair of the barbed prongs **114** inwardly to disengage the barbs from the ledge **37** and thereby allow the manifold assembly to be detached from the framework. It is noted that some of the figures depict latching members **86** in a more simplified and schematic form than shown FIGS. **12**A and **12**B which provide a more detailed view of latching member **86**.

[0070] Latching members **86** may also be referred to herein as an engagement member **86** and include a user-accessible radially enlarged head **87** that can be depressed by a user to cause camming member **112** of engagement member **86** to disengage the flexible prongs **114** and thereby allow removal of the nozzle manifold from the assembly. Other mechanisms for releasably securing manifold assembly to framework **38** may also be employed, e.g., threaded fasteners, pins and other suitable securement mechanisms.

[0071] A central opening **88** extends through framework **38** and an aligned central opening **89** is present in cover **40** to facilitate the mounting of faucet **22** on deckplate assembly **30**. FIG. **1**B provides a schematic diagram of how a faucet assembly **90** including a faucet **22** and deckplate assembly **30** are connected to a fluid supply for operation. In the illustrated embodiment, the faucet assembly **90** is connected to a hot water supply **92** and a cold water supply **94**. A fluid line **93** connects hot water supply **92** with a mixing valve **96** disposed in the body of faucet **22**. A separate fluid line **95** connects cold water supply **94** with mixing valve **96**. Faucet handle **98** is used to adjust mixing valve **96** to adjust the ratio of hot and cold water and thereby adjust the temperature of water discharged from mixing valve **96** into regulated water line **100**.

[0072] Water line **100** extends from mixing valve **100** to a diverter valve **102**. Water then flows from diverter **102** to either deck supply line **104** or faucet supply line **106**. Mixing valve **96** and diverter valve **102** are conventional valves as typically used when combining a spray head with a single handle kitchen faucet. An example of such valves is disclosed in U.S. Pat. No. 7,721,761 B2, the entire disclosure of which is hereby incorporated herein by reference. Check valves **108** are arranged in fluid lines **104**, **106** to prevent the flow of fluid from lines **104**, **106** into diverter **102**.

[0073] During operation of faucet assembly **90**, handle **98** is used to turn the water flow on and off and regulate the temperature of the water dispensed from either faucet outlet **28** or nozzles **80**. To dispense water from faucet outlet **28**, handle **98** is used to open control valve **96**, which in the illustrated embodiment is a mixing valve, and control valve **50** is closed. The relative pressures acting on the spool element in diverter valve **102** will move the spool element to a position that diverts fluid from line **100** to line **106** where it will be discharged from faucet outlet **28**. Opening valve **50** while valve **96** is also open will result in the fluid pressure moving spool element to a position where water from line **100** is diverted to line **104** where it flows to manifold assembly **60** and is discharged from the plurality of nozzles in a waterfall discharge.

[0074] In an alternative embodiment, instead of using a control valve **50** that only opens and closes to allow or deny the passage of fluid therethrough, a mixing valve could be used to provide the control valve for the manifold assembly **60**. In such an embodiment, both hot and cold water lines would be connected to the mixing valve forming control valve **50**. Both hot and cold water lines would also be connected to the mixing valve in the faucet **22** whereby water could be simultaneously dispensed from both faucet outlet **28** and nozzles **80** and the temperature of the water discharged from the faucet outlet **28** could be adjusted independently of the temperature of the water being discharged from nozzles **80**. In such an alternative embodiment, a diverter valve would not be required. The illustrative embodiment of FIGS. **15-21**, provides an example of such an alternative embodiment having a mixing valve which is supplied by both a hot water line and a cold water line.

[0075] A second illustrated embodiment of a waterfall deckplate assembly **130** is shown in FIGS. **8-10**, **12**A, **12**B and **13**. The waterfall deckplate assembly **130** includes many similar features as the waterfall deckplate assembly **30** detailed above. As such, in the following description duplicative parts are not discussed in detail and similar components are identified with like reference numbers.

[0076] This waterfall deckplate assembly **130** is similar to the first embodiment assembly **30**, but the framework **138** of the second embodiment has a greater height to define a storage compartment **132** for a spare manifold assembly **160**A and the manifold assembly **160** that is in fluid communication with fluid line **104** is positioned at a greater height above the horizontal support surface. The spare manifold may be identical to manifold **160** and act simply as a spare part in case manifold assembly **160** fails, or, it may include nozzles having a different spray pattern whereby the two manifolds can be interchangeably installed in the deckplate assembly to provide the desired spray pattern. As mentioned earlier, manifold assemblies **160**, **160**A are identical to manifold assembly **60** except for the addition of a pair of brackets **174** on the rear wall of manifold housing **172**.

[0077] Framework **138** has a bottom lip **139** on which spare manifold assembly **160**A is positioned. Cover plate **134** is then positioned to trap manifold assembly **160**A on lip **139** with the lower section **136** of cover plate **134** covering assembly **160**A. Cover plate **134** also defines a horizontal shelf surface **140** on which manifold assembly **160** is mounted. Both manifold assembly **160** and **160**A have the same design as manifold assembly **60** and manifold assembly **160** is coupled with extension **52** to communicate fluid to the internal passage of manifold assembly **160** while it is mounted on shelf **140**.

[0078] Cover plate **134** also includes back wall sections **144** located proximate the opposite ends of cover plate **134** and openings **146** in back wall sections **144** allow extension **52** to extend therethrough to engage manifold assembly **160**. Cover plate **134** also includes more centrally located back wall sections **142** that are inserted into brackets **174** on the manifold housing **172** of manifold assembly **160** to thereby help secure cover plate **134** on framework **138**. Gaps **143** are formed between back wall sections **142** and **144** to allow barbed prongs **114** to engage framework **138**. By depressing latching members **86** to release barbed prongs **114** from framework **138** and disengaging manifold assembly **160** from extension **52**, manifold assembly **160** and cover plate **134** can be removed from framework **138**. Once manifold assembly **160** and cover plate **134** are removed, the spare manifold assembly **160**A can be removed from storage compartment **132**.

[0079] Turning to FIG. **14**, a plurality of interchangeable knobs **64** and **64**A-**64**D that can be mounted on a control valve **50** are shown. Each of these plurality of user operable actuators have a different exterior appearance whereby an appearance that matches the faucet associated with the deckplate assembly can be chosen.

[0080] FIGS. **15-23** illustrate another embodiment of a faucet assembly **220**. The faucet assembly **220** of this illustrative embodiment includes a faucet **22** and a waterfall deckplate assembly **230**. While waterfall deckplate assemblies **30**, **130** of FIGS. **1**A and **8** include deckplate structures defining an opening **88**, **89** with the faucet **22** being mountable on the deckplate structure at the opening **88**, **89**, waterfall deckplate assembly **230** is adapted to be mounted proximate faucet **22** whereby both waterfall deckplate assembly **230** and faucet **22** discharge fluid into a common basin **234** with the faucet **22** being mounted separately and spaced apart from the deckplate assembly **230** as shown in FIG. **15**.

[0081] It is noted that the construction and function of waterfall deckplate assembly **230** is generally the same as assemblies **30**, **130**. As such, in the following description duplicative parts are not discussed in detail and similar components are identified with like reference numbers.

[0082] In the illustrative embodiment of FIG. **15**, both faucet **22** and waterfall deckplate assembly **230** are mounted directly on sink deck or rim **232** at locations where an opening is located in the sink rim **232**, whereby water lines **93**, **95** and **293**, **295** may be routed separately to faucet **22** and waterfall deckplate assembly **230**, respectively, from underneath sink basin **234** through openings in sink rim **232**. In this regard, it is noted that many standard sink fixtures include both faucet openings and accessory (e.g., side sprayer or sprayhead) openings. Faucet **22** may be mounted using the standard faucet openings, while waterfall deckplate assembly **230** can be mounted over an opening adapted to receive a conventional side sprayer or sprayhead and thereby replace the conventional side sprayer.

[0083] Waterfall deckplate assembly **230** functions similarly to waterfall deckplate assemblies **30**, **130** discussed above and includes a knob **64** for manipulating a user operable control valve **250** which is operable to selectively allow fluid flow to the discharge nozzles **280** of manifold assembly **260**.

[0084] In contrast to the embodiment of FIG. **1**A, waterfall deckplate assembly **230** shown in FIG. **15** is not connected downstream of a diverter valve **102**, but instead includes the control valve **250** that functions as a mixing valve and is connected to both the hot water supply **92** and the cold water supply **94**.

[0085] As can be seen in FIG. **15**, T-fittings **292**, **294** are respectively connected to the hot water supply **92** and cold water supply **94**. Water line **93** supplies hot water from T-fitting **292** to the mixing valve **96** of faucet **22**, while water line **95** connected to T-fitting **294** supplies cold water to the mixing valve **96** of faucet **22**. A faucet supply line **206** runs from the mixing valve **96** of faucet **22** to the discharge outlet **28** of faucet **22**. The extra length of supply line **206** allows the sprayhead **29** of faucet **22** to be detached from its support clip or nest, as will be understood by those having ordinary skill in the art.

[0086] A hot water supply line **293** and T-fitting **292** couple hot water supply **92** to waterfall deckplate assembly **230**, while a cold water supply line **295** and T-fitting **294** couple cold water supply **94** to waterfall deckplate assembly **230**.

[0087] As can be seen in FIG. **20**, hot water supply line **293** and cold water supply line **295** extend through the hollow body of attachment post **242** and are attached to inlet bores **253** in lower valve member **252**. An upper valve member **254** is engaged with lower valve member **252** and is rotated relative to lower valve member **252** by rotating a knob **64** attached to the stem **258** of upper valve member **254**. The mixing valve **250** is illustratively supported within a base or framework **238**. A cover **240** receives, and is positioned above, the framework **238**. A retainer, such as a mounting nut **241**, may secure the mixing valve **250** within an opening **243** of the framework **238**.

[0088] As upper valve member **254** is rotated relative to lower valve member **252**, passages **257** in upper valve member **254** act to open and close the inlet bores **253** in lower valve member **252** and regulate the flow of water to outlet passage **259** in lower valve member **252**.

[0089] It is noted that mixing valve **250** is depicted in simplified form and a wide variety of different mixing valves (including cycling valves) well-known in the art can be used with waterfall deckplate assembly **230**. For example, the mixing valves disclosed in U.S. Pat. No. 7,753,074 to Rosko et al., U.S. Pat. No. 8,469,056 to Marty et al. and U.S. Pat. No. 8,375,990 to Veros, the disclosures of which are hereby incorporated herein by reference, could be adapted for use with waterfall deckplate assemblies disclosed herein.

[0090] As those having ordinary skill in the art will recognize, a mixing valve such as valve **250** can be used to stop and start water flow through the valve, and regulate the temperature and/or flow rate of the water flow discharged from the valve when the valve is connected to both hot water source **92** and cold water source **94**.

[0091] Valve **250** is part of the fluid conveyance system **231** of assembly **230** and from valve **250**, the discharged water flows through a lateral extension **262** and spindle connector **264** to manifold assembly **260** where the water is discharged through a plurality of nozzles **280**. O-rings and flat sealing washers are illustratively used to prevent the leakage of fluid in assembly **230** as will be readily understood by those with ordinary skill in the art.

[0092] Manifold assembly **260** is secured to waterfall deckplate assembly **230** with latching members **286**. Latching members **286** are similar to and function similar to latching members **86** however, latching members **286** are represented in simplified form in the figures and to further understand how latching members **286** releasably secure manifold assembly **260** reference should be made to latching members **86** as depicted in FIG. **12**B and as discussed above.

[0093] In alternative embodiments, waterfall deckplate assembly **230** can be attached downstream of a diverter valve similar to assembly **30** as shown in FIG. **1**A. When used with a diverter valve, mixing valve **250** can be replaced with a ball valve or other form of valve that simply stops and starts the flow of fluid therethrough.

[0094] The smaller size of deckplate assembly **230** with its mounting location being separate from and spaced apart from faucet **22** provides certain advantages. For example, it allows a homeowner having an installed faucet with separate sprayhead to remove the sprayhead and replace it with a waterfall deckplate assembly **230** without having to replace or even detach and reinstall their existing faucet. This facilitates the easy and convenient upgrading of a faucet/sprayhead combination into a faucet and waterfall combination.

[0095] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the spirit and scope of the invention as described and defined in the following claims.

Claims

1. A waterfall deckplate assembly comprising: a deckplate structure adapted to be mounted on a horizontal surface; and a fluid conveyance system coupled with deckplate structure, the fluid conveyance system adapted to be connected to a fluid supply and including: a plurality of discharge nozzles supported by the deckplate structure wherein the plurality of discharge nozzles are horizontally aligned when the deckplate structure is mounted on the horizontal surface; and a user operable control valve wherein the control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles.

2. The waterfall deckplate assembly of claim 1, wherein the control valve is a positioned downstream of a mixing valve adapted to be coupled with a hot water supply and a cold water supply and a diverter valve is disposed between the mixing valve and the control valve, the diverter valve being couplable with a faucet.

3. The waterfall deckplate assembly of claim 1, wherein the plurality of discharge nozzles are adjustable nozzles.

4. The waterfall deckplate assembly of claim 1, wherein the deckplate structure defines an opening adapted to receive a faucet assembly.

5. The waterfall deckplate assembly of claim 1, wherein the control valve includes a user operable actuator projecting above the deckplate structure for operating the control valve.

6. The waterfall deckplate assembly of claim 5, comprising a plurality of user operable actuators interchangeably installable on the control valve, each of the plurality of user operable actuators having a different exterior appearance.

7. The waterfall deckplate assembly of claim 1, wherein the plurality of discharge nozzles are disposed on a first nozzle manifold and the assembly includes a second plurality of discharge nozzles disposed on a second nozzle manifold, the first and second nozzle manifolds being interchangeably installable in the fluid conveyance system; and wherein the deckplate structure defines a storage compartment whereby one of the first and second nozzle manifolds not installed in the fluid conveyance system is storable in the storage compartment.

8. The waterfall deckplate assembly of claim 1, wherein the plurality of discharge nozzles are disposed on a nozzle manifold wherein the nozzle manifold includes at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and wherein the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.

9. The waterfall deckplate assembly of claim 1, wherein the control valve is a mixing valve adapted to be coupled with a hot water supply and a cold water supply.

10. The waterfall deckplate assembly of claim 1, wherein the waterfall deckplate assembly is adapted to be mounted proximate a faucet whereby both the waterfall deckplate assembly and the faucet discharge fluid into a common basin with the faucet being mounted separately and spaced apart from the deckplate assembly.

11. A waterfall deckplate assembly comprising: a deckplate structure adapted to be mounted on a horizontal surface; a first plurality of discharge nozzles disposed on a first nozzle manifold and a second plurality of discharge nozzles disposed on a second nozzle manifold; a fluid conveyance system coupled with deckplate structure, the fluid conveyance system being couplable with a selected one of the first nozzle manifold and the second nozzle manifold, and wherein the fluid conveyance system is connectable to a fluid supply to thereby convey fluid to the selected one of the first nozzle manifold and the second nozzle manifold coupled with the fluid conveyance system; a user operable control valve disposed in the fluid conveyance system wherein the control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles; and wherein the first nozzle manifold and the second nozzle manifold are interchangeably couplable to the fluid conveyance system and the deckplate structure defines a storage compartment whereby a non-coupled one of the first nozzle manifold and the second nozzle manifold is storable in the storage compartment.

12. The waterfall deckplate assembly of claim 11, wherein the plurality of discharge nozzles of the nozzle manifold coupled with the fluid conveyance system are horizontally aligned when the deckplate structure is mounted on the horizontal surface.

13. The waterfall deckplate assembly of claim 11, wherein the storage compartment is disposed below the nozzle manifold coupled with the fluid conveyance system when the deckplate structure is mounted on the horizontal surface.

14. The waterfall deckplate assembly of claim 13, wherein the nozzle manifold coupled with the fluid conveyance system discharges fluid from a first elongate side of the deckplate structure and the storage compartment is accessible through a removable cover disposed on a second elongate side of the deckplate structure disposed opposite the first elongate side.

15. The waterfall deckplate assembly of claim 11, wherein at least one of the first plurality of discharge nozzles and the second plurality of discharge nozzles are adjustable nozzles.

16. The waterfall deckplate assembly of claim 11, wherein the first nozzle manifold and the second nozzle manifold each include at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and wherein the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.

17. A faucet assembly comprising: a faucet having a waterway connectable to a water supply, the waterway extending to a discharge outlet, and a user operable first control valve for controlling the flow of water through the waterway to the discharge outlet; and a deckplate assembly comprising: a

deckplate structure adapted to be mounted on a horizontal surface; and a fluid conveyance system coupled with deckplate structure, the fluid conveyance system adapted to be connected to a fluid supply and including: a plurality of discharge nozzles supported by the deckplate structure wherein the plurality of discharge nozzles are horizontally aligned when the deckplate structure is mounted on the horizontal surface; and a user operable second control valve wherein the second control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles independently of the operation of the first control valve.
18. The faucet assembly of claim 17, wherein the first control valve is a mixing valve adapted to be coupled with a hot water supply and a cold water supply and a diverter valve is disposed between the mixing valve and the second control valve, with both the waterway and the fluid conveyance system being disposed downstream of the diverter valve and the mixing valve being disposed upstream of the diverter valve.
19. The faucet assembly of claim 17, wherein the both the first control valve and the second control valve are mixing valves adapted to be coupled with a hot water supply and a cold water supply.
20. The faucet assembly of claim 17, wherein the second control valve includes a user operable actuator projecting above the deckplate structure for operating the second control valve.
21. The faucet assembly of claim 20, comprising a plurality of user operable actuators interchangeably installable on the second control valve, each of the plurality of user operable actuators having a different exterior appearance.
22. The faucet assembly of claim 17, wherein the plurality of discharge nozzles are disposed on a first nozzle manifold and the assembly includes a second plurality of discharge nozzles disposed on a second nozzle manifold, the first and second nozzle manifolds being interchangeably installable in the fluid conveyance system; and wherein the deckplate structure defines a storage compartment whereby one of the first and second nozzle manifolds not installed in the fluid conveyance system is storable in the storage compartment.
23. The faucet assembly of claim 17, wherein the plurality of discharge nozzles are disposed on a nozzle manifold wherein the nozzle manifold includes at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and wherein the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.
24. The faucet assembly of claim 17, wherein the deckplate structure defines an opening and the faucet is mounted on the deckplate structure at the opening.
25. The faucet assembly of claim 17, wherein the waterfall deckplate assembly is adapted to be mounted proximate the faucet whereby both the waterfall deckplate assembly and the faucet discharge fluid into a common basin with the faucet being mounted separately and spaced apart from the deckplate assembly.
26. A faucet assembly comprising: a faucet having a waterway connectable to a water supply, the waterway extending to a discharge outlet, and a user operable first control valve for controlling the flow of water through the waterway to the discharge outlet; and a deckplate assembly comprising: a deckplate structure adapted to be mounted on a horizontal surface; a first plurality of discharge nozzles disposed on a first nozzle manifold and a second plurality of discharge nozzles disposed on a second nozzle manifold; a fluid conveyance system coupled with deckplate structure, the fluid conveyance system being couplable with a selected one of the first nozzle manifold and the second nozzle manifold, and wherein the fluid conveyance system is connectable to a fluid supply to thereby convey fluid to the selected one of the first nozzle manifold and the second nozzle manifold coupled with the fluid conveyance system; a user operable control valve disposed in the fluid conveyance system wherein the control valve is operable to selectively allow fluid flow to the discharge nozzles and prevent fluid flow to the discharge nozzles; and wherein the first nozzle manifold and the second nozzle manifold are interchangeably couplable to the fluid conveyance

system and the deckplate structure defines a storage compartment whereby a non-coupled one of the first nozzle manifold and the second nozzle manifold is storable in the storage compartment.

27. The faucet assembly of claim 26, wherein the plurality of discharge nozzles of the nozzle manifold coupled with the fluid conveyance system are horizontally aligned when the deckplate structure is mounted on the horizontal surface.

28. The faucet assembly of claim 26, wherein the storage compartment is disposed below the nozzle manifold coupled with the fluid conveyance system when the deckplate structure is mounted on the horizontal surface.

29. The faucet assembly of claim 28, wherein the nozzle manifold coupled with the fluid conveyance system discharges fluid from a first elongate side of the deckplate structure and the storage compartment is accessible through a removable cover disposed on a second elongate side of the deckplate structure disposed opposite the first elongate side.

30. The faucet assembly of claim 26, wherein the first nozzle manifold and the second nozzle manifold each include at least one flexible prong for engaging the assembly and thereby securing the nozzle manifold to the assembly and wherein the assembly includes a user-accessible engagement member wherein manual manipulation of the engagement member disengages the at least one flexible prong and thereby allows removal of the nozzle manifold from the assembly.
